

Friday's Session Rough Plan:

Start with blank table: Select k objects from n choices:

	order matters	order doesn't matter
replacement Yes		
No		

Fill corresponding blank as we finish explaining formula:

Order matters, has replacement:

Select k objects from n choices:

↑ — — — —
each blank has n options,
so in total n^k ways.

Order matters, no replacement:

Select k objects from n choices:

1st 2nd 3rd ... k th.
↑ ↑ ↑ — — ↑
 n choices $n-1$ choices $n-2$ choices ... $n-k+1$ choices
 $n-1$ choices

In total of $n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot (n-k+1)$ choices.

$$n! = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot (n-k+2) \cdot (n-k+1) \cdot (n-k) \cdot \dots \cdot (3) \cdot (2) \cdot (1)$$

$$(n-k)! = (n-k) \cdot (n-k-1) \cdot \dots \cdot 3 \cdot 2 \cdot 1, \text{ therefore}$$

$$n(n-1)(n-2) \dots (n-k+1) = \frac{n!}{(n-k)!}$$

Order does not matter, no replacement:

$$\begin{array}{c} \overline{\uparrow} \quad \overline{\uparrow} \quad \overline{} \quad \overline{} \quad \overline{} \\ n \quad n-1 \quad \dots \quad \dots \quad \dots \end{array} \quad \frac{n!}{(n-k)!} \stackrel{?}{=} \text{is that it?}$$

NO! Since order does not matter, we overcount configurations!

Given a selected list of k objects, how many ways are there for us to arrange it? $\Rightarrow$ equivalent to asking:

"order matters, with no replacement, choosing k objects from k choices"

By the formula, we have $\frac{k!}{(k-k)!} = \frac{k!}{0!} = \frac{k!}{1} = k!$ ways.

So for any configuration, we overcount it for $k!$ times,

$$\frac{n!}{(n-k)!} / k! = \frac{n!}{(n-k)! k!} \stackrel{?}{=} \underline{\underline{\binom{n}{k}}}$$

Order does not matter, with replacement:

$$\begin{array}{c} \overline{} \quad \overline{} \quad \overline{} \quad \overline{} \quad \overline{} \\ n \quad n \quad n \quad \dots \end{array} \quad n^k \stackrel{?}{=} \text{is that it?}$$

NO!

Order does not matter, so we overcount.

quick guide to stars and bars:

let $X_1, X_2, \dots, X_n$ denote number of i -th choice selected.
call them X_i 's.

then $\sum_{i=1}^n X_i = X_1 + X_2 + \dots + X_n = k$, since we only have k blanks to fill in. (2)

each $X_i \geq 0$. Therefore, it is equivalent to ask for number of solutions to $\sum_{i=1}^n X_i = k$. positive integer

To see it intuitively: X_1 is number of 1's, or 1st choice of n choices
 X_2 is number of 2's, or 2nd choice of n choices
 $\vdots$
 X_n is number of n 's or n -th choice from n choices

Therefore $\sum_{i=1}^n X_i = k$. There is a more visually appealing way:

represent the k blanks as stars: $\underbrace{* * * * * \dots *}_{k \text{ of them}}$

and we are using $n+1$ bars to segment them into n different sections:

Example:

bar₁ bar₂ bar₃ bar₄ bar_{n-2} bar_{n-1}

$* \mid * * \mid * \mid \dots \mid * * * \mid * * *$

in total, it's $\$$
a string with
 $n+1+k$ symbols.

of stars on the left of i -th bar = # of selections on i -th choice from n choices,

of stars on the right of bar $n-1$ = # of selections on n -th choice from n choices

$\$$:

Therefore, we are choosing k positions from $k+n-1$ positions to be stars, and the rest can be bars.

This is essentially choosing k from $n-1+k$, order does not matter, and with no replacement:

$$\frac{(n-1+k)}{k}$$

Lecture II key points: (Ask if they are sleepy or not :))

Bayes' theorem Example:

Imagine a rare cancer that affects 0.0037% of the population.

↓
Take Glioblastoma (some brain cancer that is deadly)
for example:

Imagine a test that has following properties:

false positive rate: 1%

false negative rate: 0.1%

Now assume a patient came in and did a test on Glioblastoma, and test result is positive, what is the probability that this patient indeed has Glioblastoma?

(Ask students for their opinions)

Bayes' theorem: $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$

let $C=1$ indicate event of having cancer, $C=0$ is no cancer

$T=1$ indicate event of having positive test result, $T=0$ is negative

What we want to calculate is:

$P(C=1 | T=1)$, by Bayes' theorem we have

$$\begin{aligned}
 P(C=1 | T=1) &= \frac{P(T=1 | C=1) P(C=1)}{P(T=1)} \\
 &= \frac{P(T=1 | C=1) P(C=1)}{\sum_{i=0}^1 P(T=1 | C=i) P(C=i)} \\
 &= \frac{P(T=1 | C=1) P(C=1)}{P(T=1 | C=1) P(C=1) + P(T=1 | C=0) P(C=0)}
 \end{aligned}$$

Total probability rule

We know $P(C=1) = 0.000037$, so $P(C=0) = 1 - P(C=1)$
 by false negative rate, $= 0.999963$
 $P(T=0 | C=1) = 0.001$, so $1 - P(T=0 | C=1) = P(T=1 | C=1)$
 $= 0.999$

(let student's figure on calculation)

$$P(C=1 | T=1) \approx 0.365\% \quad (\text{Is it counter-intuitive?})$$

Expectation properties:

let X, Y be Random Variables, a, b are constants in $\mathbb{R}$.

$$E[a \cdot Y] = a \cdot E[Y]; \quad E[X + Y] = E[X] + E[Y].$$

$$\mathbb{E}[Y+b] = \mathbb{E}[Y] + b$$

Variance properties:

$$\text{Var}(aY) = a^2 \text{Var}(Y) \quad \text{Var}(Y+c) = \text{Var}(Y)$$

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2 \cdot \text{Cov}(X, Y)$$

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[Y]\mathbb{E}[X]$$

$$\text{Cov}(aX+b, cY+d) = ac \cdot \text{Cov}(X, Y)$$

(If there is still time, cover Vandermonde's identity)

$$\binom{m+n}{r} = \sum_{k=0}^r \binom{m}{k} \binom{n}{r-k}$$

Intuitively: from m left-handed students and n right-handed students, select r of them for some experiment.

equivalent to asking:

select k students from left-handed group,

$r-k$ students from right handed group.

And then range all possible values of k from 0 to r .